

Wednesday, 3rd Sept. 2025

"Statistics"

Intro + Def:

- describes how to collect data

↓
Has 2 sources

Primary

- surveys
- raw form
- unprocessed

Secondary

- processed data

- how to organize data
- how to present data
- how to analyze data
- how to interpret any numerical data or categorical data to make some decisions and draw some conclusions about the data.

Types of Statistics:

i). Descriptive Statistic:

It provides some methods for data collection, data organization and for analysis of data (using some measures).

ii). Inferential Statistic:

It provides some methods; by using these methods we can draw conclusions which help us to take decisions about population based on sample information.

∴ population:
area of interest

Data :-

Piece of information

Types of Data:-

i)- **Quantitative :-**

Any value which is measurable

ii)- **Qualitative :-**

Anything that is observable
(attributes, characteristics etc ; non-measurable)

Quantitative Types

Discrete

- countable
- no. of chairs,
no. of stools

Continuous

- measurable
- not countable
but measurable with any
instrument.
- (height, length)

Basic Terms:

i)- **Population:** Area of interest or individual possessing same characteristics.

ii)- **Sample:** Chunk of population or subset of population.

iii)- **Statistic:** Any numerical value calculated from sample of population

iv)- **Parameter:** Any numerical value calculated from population.

Friday, 5th Sept. 2025

Data Presentation / Organization

(i) - Using frequency distribution and graphs for Qualitative.

⇒ Frequency distribution is a table that shows how many times each item appear

Frequency distribution of Qualitative data:-

↳ we have to make 3 columns, when doing frequency distribution

Language used	Frequency	Relative Frequency	% F
↓ category column e.g. Eng, Urdu	↓ This shows how many times	↓ This shows proportion of each language	
↑ sum of frequency		$RF = \frac{f}{\sum F}$	

$\sum f = n$ → no. of observation

sum of frequency = number of observation

$RF = \frac{f}{\sum F}$ = proportion of data

$\%f = RF \times 100$

Graphical Presentation of Qualitative Data

↳ Pichart (circular diagram): A round chart divided into slices, each slice shows a category

↳ based on some sectors and these sectors represent the category of the data.

⇒ After making table we can draw graph to show data ∴ slice = sector

Tuesday, 9th Sept. 2025

Data Presentation:



Pie chart is a graphical representation of qualitative data which is based on subsectors and desectors, represent the category of data.

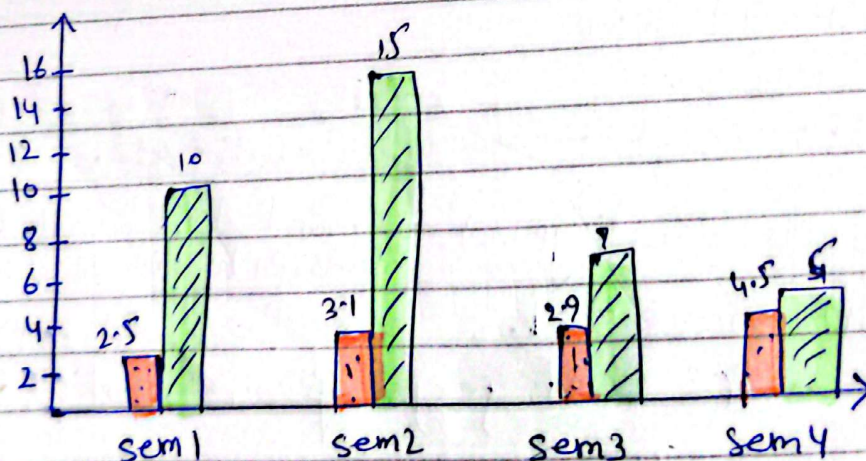
Simple Bar-Chart:

It is also a graphical form of qualitative data of frequencies. It is used for comparison, represented in the form of bars.

∴ Frequency is always whole number

Multiple Bar-Chart

Sem → Qualitative	1	2	3	4
GPA → Quantitative	2.5	3.1	2.9	4.5
No. of classes → Quantitative	10	15	7	5

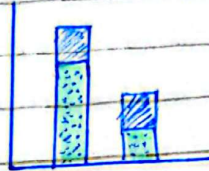


 No. of classes
 GPA

Component bar chart (task)

Also known as stacked bar chart or sub-divided bar chart represents data by dividing each bar into different segments and each segment corresponds to a component of the total value of that bar.

- 1- Actual
- 2- Percentage



Quantitative Frequency Distribution

Class interval	f	CB	CF	RF	X	%F
1-3	17	0.5-3.5	17	17/50	2	17/50 × 100
4-6	12	3.5-6.5	29	12/50	5	"
7-9	6	6.5-9.5	35	6/50	8	"
10-12	9	9.5-12.5	44	9/50	11	"
13-15	6	12.5-15.5	50	6/50	14	"
	50					

midpoint

⇒ Classes should be greater than 3 and less than 10-11

$$\text{Range} = \text{max} - \text{min} = 15 - 1 = 14$$

select no. of classes = 6

$$\frac{\text{Diff}}{\text{Interval}} = \frac{14}{6} = 2.3 \approx 2 \text{ or } 3$$

- ⇒ category's first value should fall in 1st class
- if value is discrete make frequency first.
 - if value is continuous make CB first.

→ class boundary is only used when you have to make a histogram.

⇒ Class boundary represents the exact midpoint between upper limit of one class and lower limit of next class in a frequency distribution.

Cummulative Frequency:

Running total of frequencies for a dataset showing total number of observations that fall within a certain class interval and all proceeding ones:

ones:	10	10	30	30+20+10
	20	10+20	40	30+10+20+40

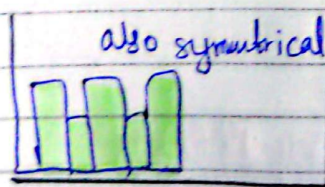
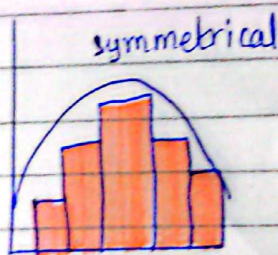
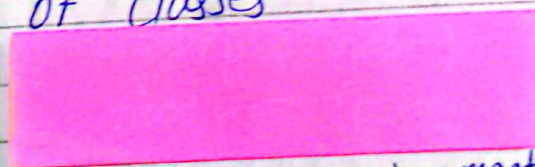
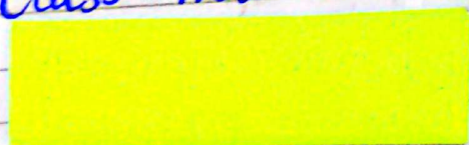
Mid Point :- (class mark) central language value of class interval in frequency distribution

$$X = \frac{\text{lower class limit} + \text{upper class limit}}{2}$$

calculated by avg of class boundary, or limit.

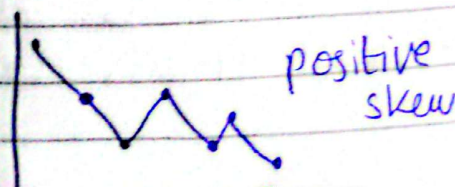
Class width:- diff b/w class boundaries

Class interval:- diff of classes



Frequency polygon:-

also a graph of frequency distribution constructed by frequency and mid point



16th Sept. 2025, Tuesday

Measure of Central Tendency

↳ a statistical value that represents the center

1. **Mean:** or typical value of a dataset.

→ sum of all values

$$\bar{X} = \frac{\sum X}{n} \rightarrow \text{number of observations}$$

(i) **Weighted Mean**

(ii) **Combined Mean**

E.g: Rent of 10 properties

5, 7, 10, 5, 7, 6, 4, 11, 10, 79

$$\bar{X} = \frac{5+7+10+5+7+6+4+11+10+79}{10}$$

$$\bar{X} = 7.2 \$ \leftarrow \text{interpret}$$

∴ **Outliers:** A number that is different from the same sequence of ranges like 2, 4, 5, 1, 2.1, 3, 2, 1.9, 2.6, 72

Q:- The Mean of 12 values is 25 after revising the values we found out that 1 value is missing by adding that value the new mean becomes 30. what is the missing value.

let outlier = x

$$x = ?$$

$$\frac{\text{sum of values}}{12} = 25$$

$$\begin{array}{l} \text{sum of} \\ \Rightarrow \text{values} = 300 \end{array}$$

$$\frac{\text{value} + x}{13} = 30$$

$$300 + x = 390$$

$$x = 390 - 300$$

$$x = 90$$

Mean (average), Mode (most frequent value), Median (middle value)

Q: A software A detects 13 outliers when we execute it 20 times and software B detects 19 outliers when we execute 30 times, what is the overall outliers detected by these softwares or detected

$$\frac{20(13) + (19)30}{50}$$

$$\frac{830}{50} = 16.6$$

Weighted Mean ::

Example 1

Items	Cost	w	xw
Food	30	4	120
Fuel	20	2	40
Clothing	40	1	40
Rent	25	3	75
Fee	70	5	350

Def of Weighted Mean :-

When each value of your dataset isn't equally important, each value has different preference or important and we have to find the average for this case property of mean is used.

17th Sept. 2025, Wednesday

Measure of Central Tendency

1. Mean

→ Weighted Mean $\rightarrow X_W = \frac{\sum W_i X_i}{\sum W_i}$
→ Combined Mean

Median:

Median is the middle value of dataset, it divides the dataset into two equal parts (after arranging data in ascending or descending order), 50% observation falls above the median and 50% below observation

$$M = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ value in data}$$

→ position

Example:-

10, 5, 7, 10, 5, 7, 6, 4, 11, 7

Arr 4, 5, 5, 6, 7, 7, 7, 10, 10, 11

$$M = \left(\frac{n+1}{2} \right)^{\text{th}} = \left(\frac{10+1}{2} \right)^{\text{th}} = 5.5^{\text{th}} \text{ value in data}$$

$$\frac{5^{\text{th}} + 6^{\text{th}}}{2} = \frac{7+7}{2} = 7$$

Median is not affected by extreme values, so it is superior than mean and appropriate when data contains outli.

Tuesday, 23rd Sept. 2025

Median calculation

3.75, 2.95, 7.11, 3.15, 3.91, 7.15, 6.51, 5.91,
6.71, 7.01, 5.51, 4.32

Arranging

example
★ 2.95, 3.15, 3.75, 3.91, 4.32, 5.51, 5.91, 6.51, 6.71,
7.01, 7.11, 7.15

$$\frac{5.51 + 5.91}{2} = \frac{11.42}{2} = 5.7$$

$$\frac{6+7}{2} = \frac{13}{2} = 6.5^{\text{th}} \text{ value}$$

$$M = \frac{n+1}{2}^{\text{th}} \text{ value}$$

$$= \frac{13}{2} = 6.5^{\text{th}}$$

$$= \frac{6^{\text{th}} + 7^{\text{th}}}{2}$$

$$= 5.7$$

- Quantiles:-

1. Quartiles
2. Deciles
3. Percentiles

Quartiles,

Quartiles divides dataset into 4 equal parts, Q_1 , (lower quartile), Q_1 contains 25% data below it and 75% above it, Q_3 (upper quartile), Q_3 contains 75% data below it and 25% data above it, and Q_2 is called Median too.

Deciles:

Deciles are the 9 values that divide the data into 10 equal parts and the values are represented by $D_1, D_2, \dots, D_9$. There is 10% data between any two points.

Percentile:

It tells you the exact position of your value or data.

$$\begin{aligned}\text{For Quartiles, } \rightarrow & 3.75^{\text{th}} \\ & 3^{\text{th}} + 0.75 (4^{\text{th}} - 3^{\text{rd}}) \\ & = 4.25\end{aligned}$$

$$4^{\text{th}} + 0.25 (5^{\text{th}} - 4^{\text{th}})$$

Quartiles:

Quartiles

$$Q_1 = \frac{1(n+1)}{4} \text{ value in data}$$

$$Q_2 = \frac{2(n+1)}{4} = \text{Med}$$

$$Q_3 = \frac{3(n+1)}{4}$$

$$\text{For example } \star \frac{1(12+1)}{4} = \frac{13}{4} = 3.25^{\text{th}} \text{ value}$$

1st quartile

3rd quartile

$$\begin{aligned}\frac{3(12+1)}{4} &= \frac{39}{4} = 9.75^{\text{th}} \text{ value} \\ &= 9^{\text{th}} + 0.75 (10^{\text{th}} - 9^{\text{th}}) \\ &= 6.935\end{aligned}$$

$$= 3^{\text{rd}} + 0.25 (4^{\text{th}} - 3^{\text{rd}})$$

$$3.75 + 0.25 (3.91 - 3.75)$$

$$= 3.79$$

MODE:-

Mode is the most frequent (repeated) value in your dataset. Mode is used as an average for qualitative data as well as for quantitative data.

Example: $5, 9, 10, 11, 15 \rightarrow$ no mode
 $7, 11, 15, 15, 9, 11, 7 \rightarrow 7, 15, 11$ are mode

Mode can be multiple and there can be no mode too

• IF TWO DATASETS HAVE SAME MEAN, MODE, MEDIAN THAT DOESN'T MEAN THE DATASETS ARE SAME, THEY CAN AND, CANNOT BE, WE CAN'T CONFIRM.

Measurement Scales:

↳ Nominal scale
Ordinal scale
Interval scale
Ratio scale } for Qualitative + Quantitative DATA

Measure of central tendency

∴ Used for data summarization

↳ HOW TO DEFINE SHAPES OF DATA/DISTRIBUTION.

1st Method: Histogram

2nd Method: Using mean, median. If mean is equal to median it means data is symmetrical. Bell shape

3rd Method: Boxplot
→ Positive skew - when mean > median
→ If mean < median, negatively skewed

Method: Boxplot

Wednesday, 24th September 2025

Measure of Central Tendency ← Descriptive

- Mean :-

∴ job outliers ni honge

- * weighted mean.
- * combined mean.

- Mode *

- Median :-

- * Quartiles
- * Deciles
- * Percentiles

∴ job outliers honge

2) Measure of Dispersion / Variation

→ Range

→ Var / SD always positive

→ IQR (Inter Quartile Range)

Range :- Max - Min

$$\text{Var}(S)^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$$

$$\text{SD}(S) = \sqrt{\text{Var}}$$

x	x ²
2	4
4	16
1	1
3	9

$$n = 4$$

$$\sum x = 10$$

$$\sum x^2 = 30$$

$$\begin{aligned}\text{var} &= \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \\ &= \frac{30}{4} - \left(\frac{10}{4}\right)^2 \\ &= \frac{30}{4} - \frac{100}{16}\end{aligned}$$

30th Sep. 2025, Tuesday

Measure of Dispersion

1) - Variance / Standard Deviation

2) - IQR = $Q_3 - Q_1$

* For comparison units need to be same.

Variance:-

$$S^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2 \quad \text{Variation}$$

→ stand deviation
SD:-

$$S = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2} \quad \text{stand deviation}$$

where n = no. of observation

$$\bar{x} = \frac{\sum x}{n}$$

Example:-

x	x^2
3	9
4	16
5	25
3	9
1	1
2	4
18	64

center is where there are

variance and standard deviation same

$$S^2 = \frac{64}{6} - \left(\frac{18}{6} \right)^2$$

$$= 10.66 - 9$$

$$\text{Variance} = 1.66$$

$$\text{Standard deviation} = \sqrt{1.66}$$

when the values are same, there is no variation
e.g. whole class has 5, the avg is 5 with 0 variation.

→ outliers x
→ Mean (standard deviation)

→ Med if outliers
→ IQR

$$\text{IQR} = Q_3 - Q_1$$

1st quartile
3rd quartile

→ if outliers or not use median for comparing datasets

∴ Data comparison and Data analysis

Class Activity

Marks of Sect A

Dataset 1

x	x ²
3	9
2	4
1	1
3	9
2	4
13	256
12	144
11	121
11	121
4	16
5	25
4	16
7	49
13 values	13 values

Marks of Sect B

Dataset 2

15	225
12	144
6	36
2	4
7	49
12	144
4	16
8	64
3	9
9	81
1	1
1	1
3	9
13	256
11	121
15 values	15 values

- R Language
- excel

Q:- How to interpret two datasets

$$\text{Mean}(\bar{X}) = \frac{78}{13} = 6$$

$$\text{Mean} \bar{X} = \frac{107}{15} = 7.13$$

$$\text{Mean}(\bar{X})^2 = \frac{775}{13} = 59.6$$

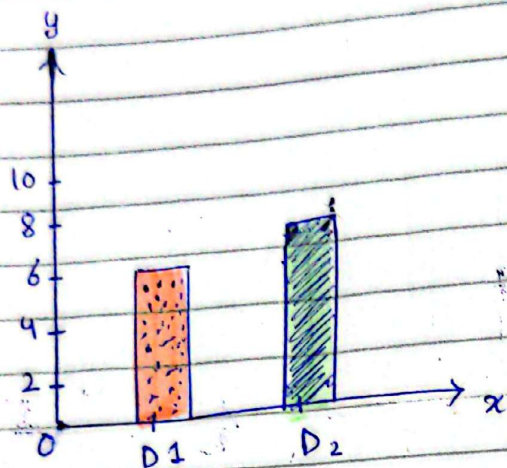
$$\text{Mean}(\bar{X})^2 = \frac{1160}{15} = 77.33$$

$$\begin{aligned} \text{Variance} &= 59.6 - 36 \\ &= 23.6 \\ &= 23.6 \end{aligned}$$

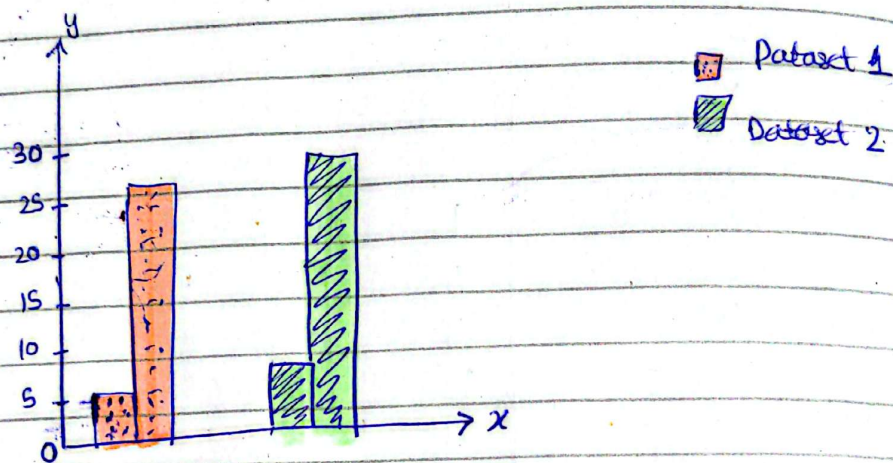
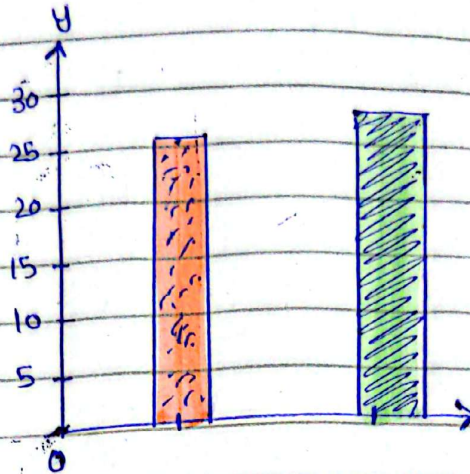
$$\begin{aligned} \text{Variance} &= 77.33 - 50.83 \\ &= 26.5 \\ &= 26.5 \end{aligned}$$

Analysis → The avg of dataset 2 is greater than dataset 1, it means dataset 2 is more superior to dataset 1, on basis of average and variation of dataset 2 is greater than dataset 1 so dataset 2 is superior.

Mean chart



Standard Deviation



→ EDA (Exploratory data analysis)

→ Coefficient of variation It is used when data has different units and we have to compare these datasets. Coefficient of variation converts the data into percentages or proportions.

The data having large coefficient of variation considered as more superior or more variant, and data having less coefficient of variation means, this data is consistent.

$$CV = \frac{S}{\bar{X}} \times 100$$

$\nearrow$ standard deviation
 $\searrow$ mean

Wed, 1st Oct. 2025

"Probability and Statistics"

- Measure of Dispersion

- ↳ Variance / Standard deviation
- ↳ IQR (inter quartile range), which describes scatterness of data
↳ $Q_3 - Q_1$

- Measure of central Tendency

1 - Mean

- ↳ Combined Mean
- ↳ Weighted Mean

Outlier:

$$\text{Lower fence} = Q_1 - 1.5(IQR)$$

2 - Median

- ↳ Quartiles
- ↳ Deciles
- ↳ Percentiles

$$\text{Upper fence} = Q_3 + 1.5(IQR)$$

3 - Mode

↗ also called
five

Exploratory Data Analysis:-

It defines

basic features of data and contains:

1 - Min

2 - Q_1

3 - Median

4 - Q_3

5 - Max

Descriptive
Statistics

Graphical Representation

BOX-PLOT

Descriptive Statistics

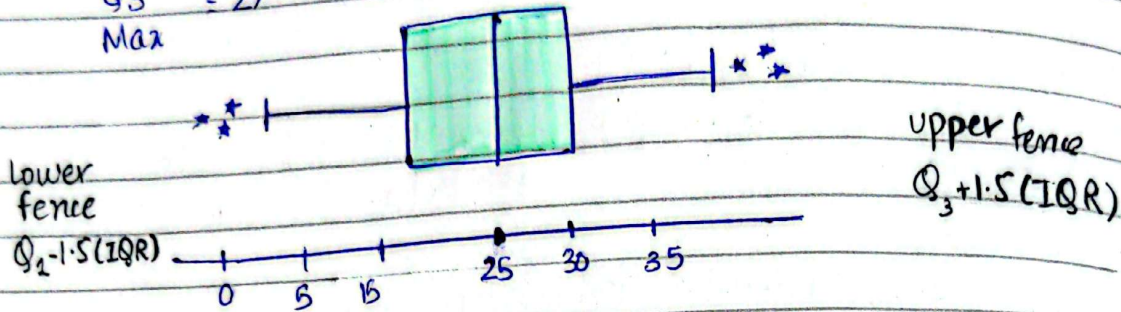
Min

$$Q_1 = 19$$

$$\text{Median} = 25$$

$$Q_3 = 29$$

Max



Revision

- Mean vs Standard deviation
- Coefficient of variation

When comparing or summarizing two datasets use medians if there are outliers, and if outlier ^{of both} than mean.

after median IQR is most,

when describe → then EDA

when summarize → compare mean / median accordingly

- when of same dataset, two types have different units like watt for data 1, and Hz for data 2, use coefficient of variation not variance formula, which is

$$\text{coefficient of variation} = \frac{\sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}}{\text{mean}} \times 100$$

Wed, 22nd Oct. 2025

Types of Events

1- Simple Event :-

An event consisting only one element is called simple event.

Example: $S = \{1, 2, 3, 4, 5, 6\} \rightarrow A = \{\text{multiple of } 5\}$
 $S = \{5\} \quad A = 5 \rightarrow \text{simple}$

2- Compound event :-

An event consisting of more than one element.

Example: $S = \{\text{even no.}\}$

$\downarrow$
 $B = \{2, 4, 6\} \rightarrow \text{compound event}$

Absolute Measure

Relative Measure